

Exercise on Producer theory- Answers

Question 1:

Assume a firm with the following production function:

TP (Q)	L	MP	FC	Cost(L)	VC	TC	MC	ATC	AVC
0	0		80	10	0	80			
1	8	0.125	80	10	80	160	80	160	80
2	12	0.25	80	10	120	200	40	100	60
3	17	0.2	80	10	170	250	50	83.3	56.7
4	23	0.17	80	10	230	310	60	77.5	57.5
5	30	0.14	80	10	300	380	70	76	60
6	38	0.125	80	10	380	460	80	76.7	63.3
7	48	0.1	80	10	480	560	100	80	68.5

It is known that one unit of labor (L) costs 10 USD.

1. Complete the table.
2. What is the minimal price that the firm will produce in the short term?
3. What is the minimal price that the firm will produce in the long term?
4. It is known that the **price is 70 USD**. What would be the production of the firm?
5. What are firm's profits in the short run?
6. What are firm's profits in the long run?
7. Will the firm produce in the short run?
8. Will the firm produce in the long run?

Answers:

2. $\min AVC = 56.7$. Below that price the firm will not produce in the **short term**
3. $\min ATC = 76$. Below that price the firm will not produce in the **long term**
4. The rule of production is $MC = P$. Therefore, if $P = 70$, the firm will produce 5 units of Q.
5. **Profits** (short-run) = $P \cdot Q - VC = 70 \cdot 5 - 300 = \underline{50}$

In the short run, fixed costs are sunk costs, because they are in the past and cannot be changed. Therefore, they play no role in economic decisions about future production or pricing.

6. **Profits** (long-run) = $P \cdot Q - TC = 70 \cdot 5 - \underline{380} = \underline{-30}$

All costs are variables and therefore all the costs (fixed and variable) have to be taken into account

7. Since in the short run the profits are positive, the firm will produce.
8. Since in the long run the profits are negative, the firm will not produce.

Question 2:

There is a market with **8 workers** who should be allocated between two fields with the following production function:

Field A

L	TP
0	0
1	5
2	9
3	12
4	14
5	15

Field B

L	TP
0	0
1	3
2	7
3	10
4	12
5	13

- How many workers would work on each field if it is known that the price of one produced unit on field A is **2\$** and the price of one produced unit on field B is **1\$**?
- What would be the salary of each worker?
- What would be the profit of the owners of each field?

Answers:

Field A

L	TP	MP (Marginal Product)	Price (\$)	VMP (Value Marginal Product)	ΣVMP
0	0		2		
1	5	5	2	10	10
2	9	4	2	8	18
3	12	3	2	6	24
4	14	2	2	4	28
5	15	1	2	2	30

Field B

L	TP	MP (Marginal Product)	Price (\$)	VMP (Value Marginal Product)	ΣVMP
0	0		1		
1	3	3	1	3	3
2	7	4	1	4	7
3	10	3	1	3	10
4	12	2	1	2	12
5	13	1	1	1	13

- There will be 4 workers in Field A and 4 workers in field B $\rightarrow 28+12 = \mathbf{40}$ (or 5 workers on field A and 3 workers on field B $\rightarrow 30+10=\mathbf{40}$).
- Profit is maximized when the VMP for the last worker hired equals the wage rate: **VMP = W**. Therefore, **W = 2**.
- $\text{Profit(A with 4 workers)} = TP \cdot p - W \cdot L = 14 \cdot 2 - 2 \cdot 4 = 20$
 $\text{Profit(A with 5 workers)} = TP \cdot p - W \cdot L = 15 \cdot 2 - 2 \cdot 5 = 20$
 $\text{Profit(B with 4 workers)} = TP \cdot p - W \cdot L = 12 \cdot 1 - 2 \cdot 4 = 4$
 $\text{Profit(B with 3 workers)} = TP \cdot p - W \cdot L = 10 \cdot 1 - 2 \cdot 3 = 4$